

Rajarshi Banerjee

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Skills

Languages: Python, Java, C#, C/C++, R, SQL

Technologies & Tools: PyTorch, Tensorflow, HuggingFace, DSPy, H2O.ai, Teradata, GCP, Spark, PySpark, Kubernetes, Docker, Splunk, Arduino, Sphinx4

Work Experience

Freelance , Remote Work

Aug 2024 - Present

Data Analyst

- Leveraged LLMs with RAG to accurately extract and validate ICD and CPT codes from unstructured medical documents, supporting automated adjudication and analytics for customer health insurance claims.
- Used DSPy framework to perform prompt tuning using the GEPA algorithm as a alternative to finetuning open source LLMs for the purposes of data extraction from medical documents.
- Leveraged Python frameworks (Pandas, NumPy, Scikit-learn) to automate ETL workflows and ensure robust data pipelines.
- Conducted statistical analyses (regression, A/B testing, time series forecasting) to drive insights from sales, website, and campaign data, translating creative metrics into quantifiable KPIs.

Wells Fargo, Bangalore

Jan 2023 - July 2024

Senior Quantitative Analyst

- Built a machine learning framework for training unsupervised HTTP anomaly detection models for bank customers who use third party banking apps by utilizing data stored in enterprise datalakes queried using Splunk. The process involved rigorous data analysis and for feature engineering and automation of the model selection process using PySpark pipeline and H2O complete with debug capabilities over large-scale systems. The project is set to be deployed at the end of the year with a revenue generation potential of more than 100 million dollars. This is currently being extended to a project to detect anomalies in Docker containers orchestrated by Kubernetes clusters.
- Built a RAG-based LLM pipeline to answer analyst queries using internal financial reports and regulatory documents, combining vector search with prompt grounding to ensure accurate, source-backed responses. Reduced manual document review time and improved compliance by restricting outputs to approved data sources.

Wells Fargo, Bangalore

Jan 2021 - Jan 2023

Quantitative Analyst

- Operated as the key technical member in a collaboration between Indian Institute of Science (IISc) and Wells Fargo to build a generative AI for synthetic customer complaint generation and was responsible with collaborating, supervising and validating the model delivered. The model is planed to be used in various line of businesses and other downstream in-house models.
- Extended a previously created third party vendor model verification pipeline to incorporate PySpark for greater scalability and better optimization resulting in over 10 million dollar revenue generation.

Wells Fargo, Bangalore

Aug 2020 - Jan 2021

Program Associate

- Developed a third party vendor model verification pipeline where various third party black box vendor models used by the company would be validated. The pipeline involved creating interfaces for accessing multiple databases and implementing functions using Pandas and Sklearn for the various metrics required by the Line of Business to evaluate their efficacy.

Education

Indian Institute of Science

Aug 2017 - Jun 2020

Master of Computer Science and Engineering, Department of Computer Science and Automation

Relevant Coursework: Data Structures and Algorithms, Computational Methods of Optimization, Artificial Intelligence, Machine Learning, Deep Learning, Reinforcement Learning, Image Processing, Natural Language Processing

Indian Statistical Institute

June 2016 - July 2016

Cryptography Summer Intern

Relevant Coursework: Quantum Computing, Quantum Cryptography

Publications

Learning Skills to Navigate without a Master: A Sequential Multi-Policy Reinforcement Learning Algorithm
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2022 ([arxiv](#)) - My research thesis where we propose a novel reinforcement learning algorithm to continuously learn new policies as per the requirement of the environment. ([video](#))

Project Work

- **3D Car-Simulator for Reinforcement Learning** ([code](#)) - Created a 3D simulated environment with the Panda3D game engine to simulate a car in various complex custom made environments. This was wrapped in Open-AI's gym to allow bench-marking of the environment on various state-of-the-art reinforcement learning algorithms. For ease of usage it was built as a docker image so that it can easily be containerized and deployed on any machine.
- **Novel Resnet Few Shot Learning Architecture** - Proposed various architectural enhancements to improve the accuracy of few learning architecture on the Omniglot and Mini-Imagenet data sets. The proposed architecture used multiple residual networks and skip connections to achieve a 5-way 1-shot accuracy of 50% on the min-Imagenet dataset, slightly lower than the state-of-the-art methodologies at the time.
- **Detailed analysis of various GAN architectures** ([code](#)) - In this project we analyzed the different variants of Generative Adversarial Networks (GANs), more specifically implementing Deep Convolution GANs (DCGANs) and Wasserstein GANs (WGANs) on MNIST and LSUN data sets and report the hyper-parameters best responsible for optimal results using the PyTorch framework.
- **Automata Theory GUI** ([code](#)) - A simple project to visualize various concepts of Automata Theory and Theory of Computation in an interactive GUI using C#. It was designed, developed and tested with custom ui design to make it an accessible technology for teachers.
- **Voice controlled bot** - A college project where I built a vehicular bot using an Arduino leveraging the Sphinx4 library to control the bot using voice commands.

Awards and Certificates

- Secured AIR 579 in GATE Computer Science & Technology 2017
- Gave a talk on deep learning for the CSA Summer School 2018, introducing the interns to the basics of deep learning and how it can be a valuable asset for a Software Engineer as well.
- Top ranked competitive coder ranking in the top 6% worldwide in Leetcode having acquired the prestigious knight badge.